Use of Galaxy tools for microbiome analysis of breast cancer RNA-seq transcriptomic data

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Introduction

In recent years, scientists were discovering presence of microoraganisms in various body sites and their possible effect on diseases. Here we present our work on this topic and posibilities of microbial identification from RNA-seq transcriptomic data using tools in Galaxy. We studied RNA-seq data from 23 tissue samples, 18 collected from primary breast tumors and 5 collected from normal breast tissue. Additionally we performed the same analysis on SRA data PRJNA553096, originated in China to compare with our own data.

Methods and study design

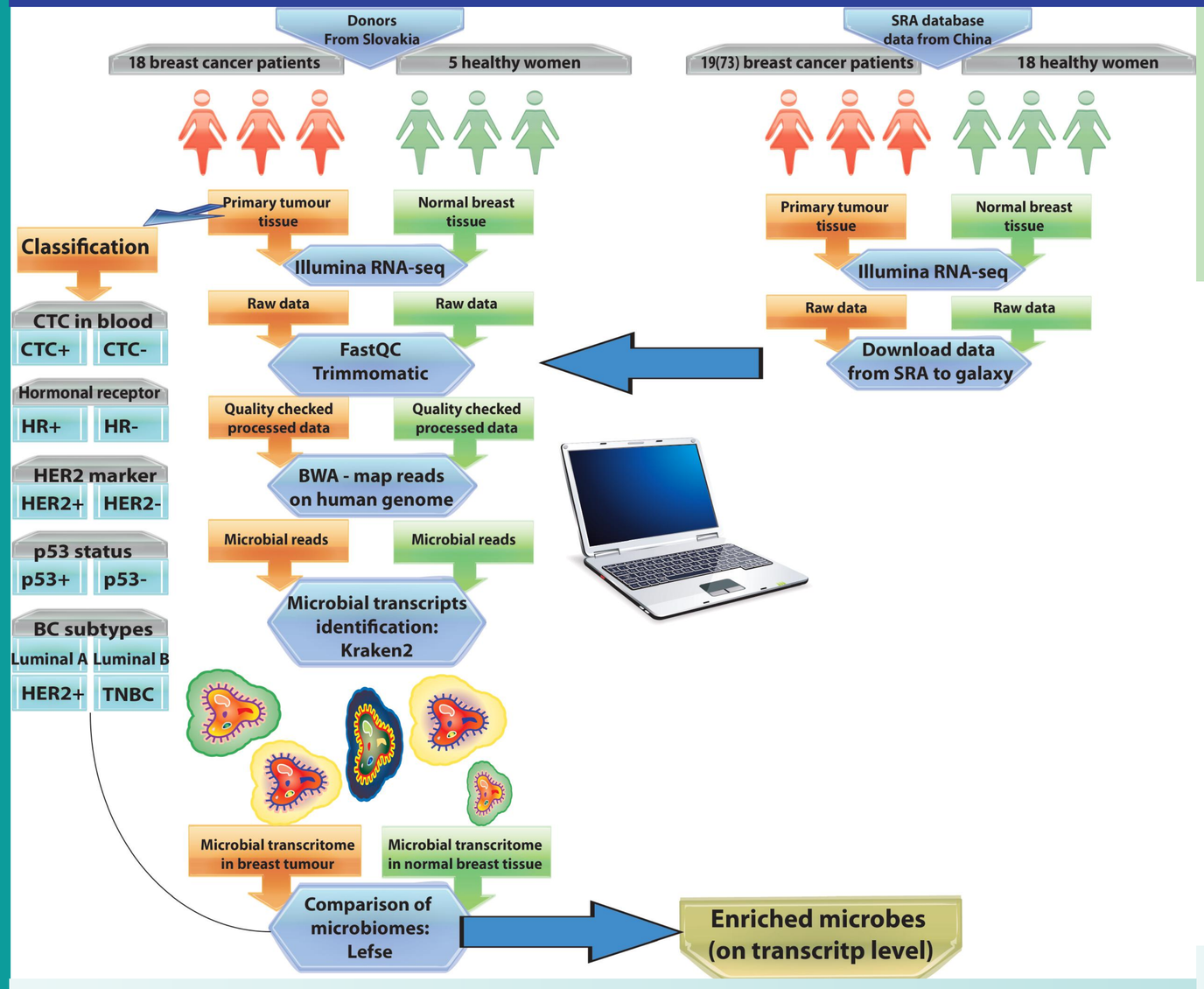

Figure 1: Design of the experiment. From donors, through experimental phase to computational analysis in Galaxy. Additional Galaxy tools used: Samtools view (extraction of unmapped reads), SamToFastq and Krona pie chart (for visualization).

Results

The most abundant species in breast tissue were *Proteobacteria*, then *Firmicutes* and *Actinobacteria* for both datasets. In Slovak samples also *Bacteroidetes* were abundant, while in Chinese samples *Cyanobacteria* were plentiful instead. Although ratios were slightly changed in tumour tissues, same phyla remained prevalent. We compared amounts of microbial transcripts between breast tumours and normal breast tissue of healthy patients by LefSe and we did the same for multiple disease groups (clasified by markers). We observed vaious microbes, which correlated with disease phenotypes. It appaers, tumour tisue harbor less rich microbiome then normal breast tissue.

Acknowledgements

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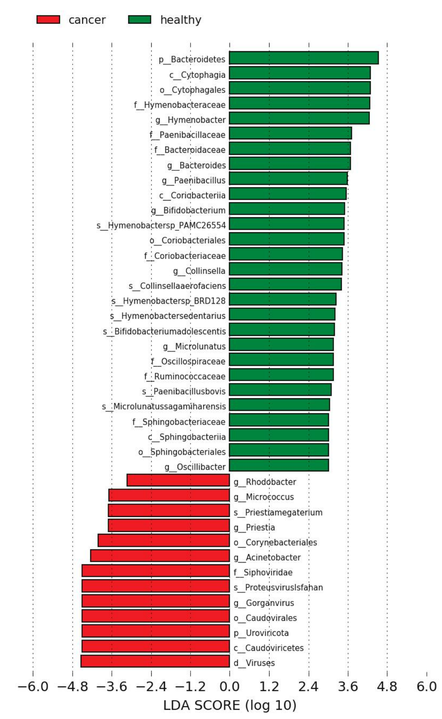
Conclusions

We confirmed results of previous studies, that identified the same prevalent phyla occupying normal breast tissue or breast tumour tissue. We also observed microbes correlated with cancer or cancer phenotypes. We succesfully tested performence of our pipeline for microbial identification by aplying it on artificial datataset, constructed from microbial genomes by ARTillumina.



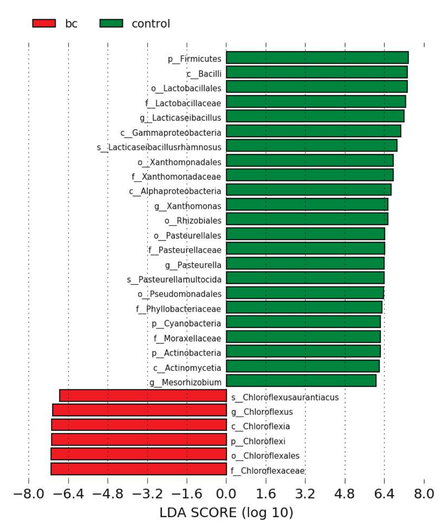
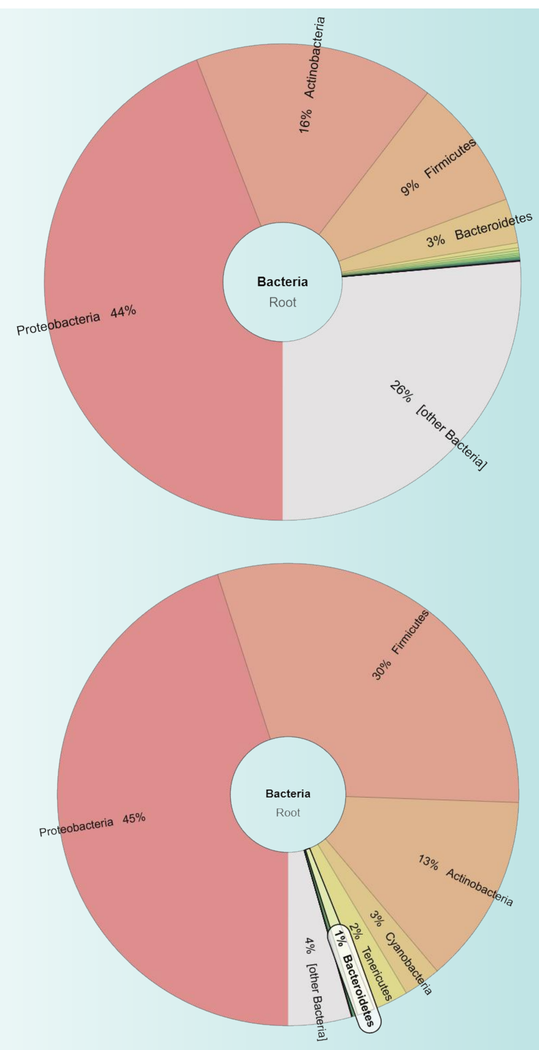


Figure 2: LefSe analysis: Microbes enriched in breast tumour (red) or normal breast tissue (green). A. Our samples from Slovakia (left) B. SRA data from China, with stricter threshold(right)



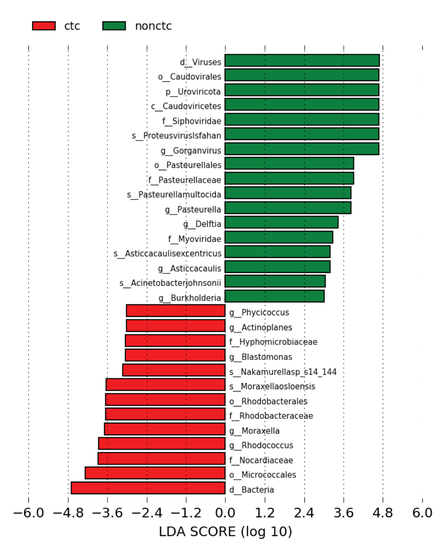


Figure 3: LefSe analysis: Microbes enriched in tumours of patients with circulating tumour cells (CTC) in their blood (red) and those enriched in patients without CTC (green).

Figure 4: Krona graph. Prevalent microbial phyla of breast tumours. A. Patients from Slovakia. B. Patients from China